

SUBJECT	Lower Secondary Science Year 8
TOPIC	Health and Lifestyle
ASSIGNMENT	One
STUDENT NAME	

FEEDBACK AND REFLECTION:

As this is your first assignment, you do not have feedback from previous work to consult before you begin. But in future assignments, we will encourage you to remind yourself of the feedback on your previous piece of work. Why do you think this will be important? Tick all that apply to you:

- ☐ I will be able to build on strengths in my work that my tutor has identified.
- ☐ I will avoid making the same mistakes that I made before.
- ☐ I will have a clearer idea of what I need to aim for in my work.
- ☐ I will feel more confident about my work.
- ☐ I will be able to act on my tutor's advice and so make progress.

What matters most to you in the feedback you receive from your tutor? Let us know here:

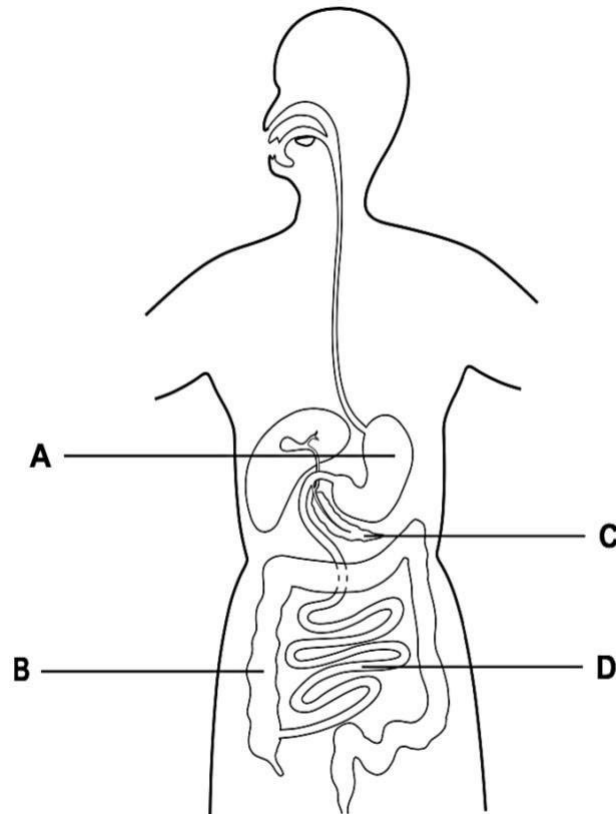
ASSIGNMENT GUIDANCE:

- You should complete this assignment by printing out this sheet and then handwriting in black ink, unless you have obtained specific permission to submit typed work. Then scan it in and submit it, as a single document.
- This is not a timed assignment. We estimate that this assignment may take you between 40 and 50 minutes for the assignment itself. However, if you need to take longer, then please do so.
- Make sure you write your full name clearly in the box at the top of this page.
- In order to complete this assignment, you will need a calculator, ruler, your coursebook and notes.

LS Science Assignment One

HEALTH AND LIFESTYLE

- The diagram shows the digestive system.



- Draw a line from each **letter** to the correct **organ** and its **function** in the digestive system. One has been done for you.

organ	letter	function
large intestine	A	absorbs water
stomach	B	digests protein
small intestine	C	produces enzymes
pancreas	D	absorbs digested food

(3 marks)

b. Some athletes take glucose tablets before a race. Why do they take glucose?

Tick the correct box.

- for growth
- for healthy bones and teeth
- to prevent disease
- to provide energy

(1 mark)

c. The table below shows what four people ate for lunch.

name	lunch
Jon	chicken and salad
Nadia	cheeseburger and chips
Clare	lemonade and a jam doughnut
Zak	mushroom soup and an orange

i. Whose lunch had the most sugar in it?

(1 mark)

ii. Whose lunch had the most fat in it?

_____.

(1 mark)

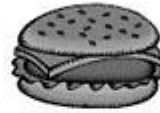
iii. Eating too much fat is bad for you. Give one reason for this.

(1 mark)

(Total 7 marks)

2. The card shows the amounts of fat and fibre in some types of food and drink from a café.

<i>Chez Jean Café</i>		
	fat in g	fibre in g
<u>type of burger</u>		
single burger	17	0.8
double burger	38	1.2
cheeseburger	21	1.0
<u>type of drink</u>		
strawberry milkshake	8	0
medium cola	0	0
medium orange juice	0	0
<u>type of potato</u>		
regular fries	15	3
baked potato	0	9



a. From the card above, choose a meal consisting of a burger, a drink and some potato, to give:

- the least fat;
- the most fibre.

Write your answers in the table below.

Food and	meal with the least fat	meal with the most fibre
type of burger		
type of drink		the drinks do not contain fibre
type of potato		

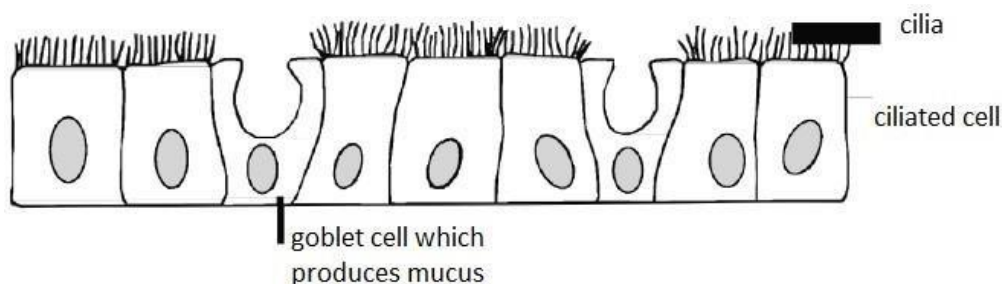
(2 marks)

b. Draw a line from each nutrient to the main reason why it is needed. Draw only four lines.

nutrient	main reason why the nutrient is needed
calcium	to keep the intestine working properly
fibre	for healthy teeth and bones
protein	for insulation
sugar	to provide energy
	for growth and repair

(4 marks)
(Total 6 marks)

3. The diagram shows two types of cell in the lining of the windpipe.



a.

i. These cells work together to keep the lungs free of bacteria and dust particles. What word describes a group of similar cells which work together?

(1 mark)

ii. Mucus is a sticky substance.

Describe how mucus and cilia keep the lungs free of bacteria and dust particles.

(2 marks)

- b. When a person breathes in cigarette smoke, the goblet cells produce extra mucus and the cilia are damaged.

What will be the consequences of this?

(2 marks)

- c. Give the names of two harmful substances in cigarette smoke. In what way is each one harmful?

i. name of substance _____

harmful effect _____

ii. name of substance _____

harmful effect _____

(4 marks)

(Total 9 marks)

4. Some people have unbalanced diets. This can give them problems with their health.

a. Draw one line from each unbalanced diet to the health problem it can cause.

unbalanced diet	health problem
too much fat	not much energy
too much sugar	heart disease
not enough protein	tooth decay
not enough carbohydrate	poor growth

(4 marks)

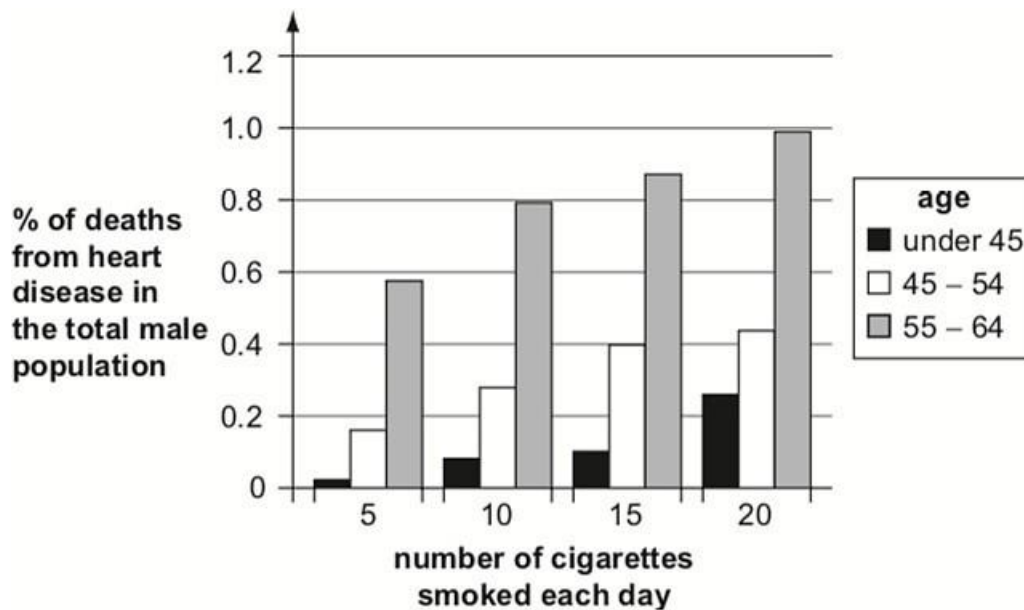
b. Explain why eating a lot of chips every day is bad for your health.

(1 mark)

(Total 5 marks)

5.

The bar chart below shows how the number of cigarettes smoked is linked to the percentage of deaths from heart disease in the total male population.



- a. Use the information in the bar chart to write **two** conclusions about the relationship between smoking and the number of male deaths from heart disease.

1. _____
- _____
2. _____
- _____

(2 marks)

- b. Smoking can cause fat to be deposited in the arteries to the heart muscle. Explain how this could prevent the heart muscle from working properly.

(3 marks)

(Total 5 marks)

6.

a. Pineapple juice contains a substance that speeds up the digestion of protein.

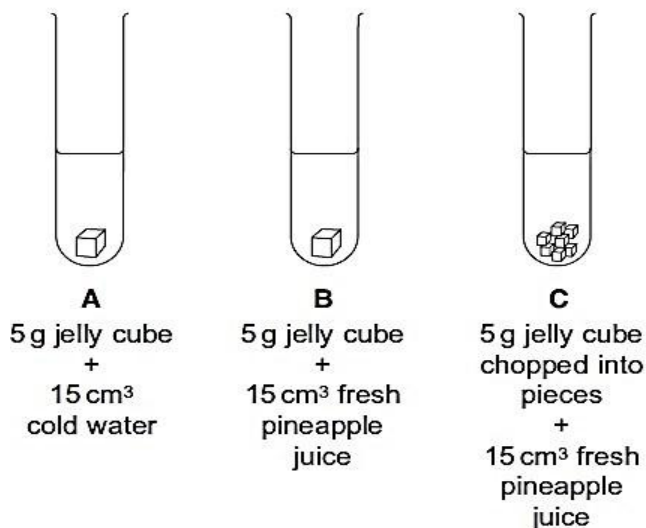
i. What is the name for substances that speed up digestion?

(1 mark)

ii. What happens to a molecule of protein during digestion?

(1 mark)

b. Asim did an experiment to investigate the digestion of gelatin. Gelatin is the protein in jelly. In test tubes A and B he used one cube of jelly in each. In test tube C he used one cube of jelly that he had chopped up.



He recorded how long it took for the jelly to be digested in each test tube. The table below shows his results.

test tube	result
A	not digested after 2 hours
B	jelly digested in 2 hours
C	jelly digested in 1 hour

i. What was the purpose of test tube A?

(1 mark)

- ii. It is helpful to chew your food. How do the results in test tube C show this?

(1 mark)

- c. The substances that speed up digestion stop working when they have been boiled.

- i. What does Asim need to put in a fourth test tube to test this in his experiment? Label test tube D. Test tubes A, B and C contain the same as in the first experiment.

			
A	B	C	D
5 g jelly cube	5 g jelly cube	5 g jelly cube	
+	+	chopped into	
15 cm ³ cold water	15 cm ³ fresh pineapple juice	pieces	
		+	
		15 cm ³ fresh pineapple juice	

(2 marks)

- ii. Predict what Asim would observe in test tube D after 2 hours.

(1 mark)

(Total 7 marks)

TOTAL FOR ASSIGNMENT 39 MARKS